

# **ESAT CAMP MOCK B**

Full sitting

EC-FULL-B

**MATHEMATICS 1**

**40 MINUTES**

**MATHEMATICS 2**

**40 MINUTES**

**PHYSICS**

**40 MINUTES**

**CHEMISTRY**

**40 MINUTES**

**BIOLOGY**

**40 MINUTES**

## **INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but **do not open this question paper until you are ready to begin**.

This paper contains **5 modules** (Mathematics 1, Mathematics 2, Physics, Chemistry, Biology). Each module has **27 multiple-choice questions** and is designed for **40 minutes** (about **200 minutes** in total if you sit every module).

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

**Please wait until you are ready before turning this page.**

*Original ESAT CAMP practice material. Not an official UAT-UK or Pearson paper.*

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## **Mathematics 1**

- 1 A shop sells two items, item P and item Q.

In January, the price of item P was 25% higher than the price of item Q.

In February, the price of item P decreased by 20%, while the price of item Q increased by 10%.

In March, the price of item P increased by 15%, while the price of item Q decreased by 5%.

In March, the price of item P was £6.30 more than the price of item Q.

What was the price of item Q in January?

- A £42
- B £50
- C £60
- D £66
- E £70
- F £75
- G £600

- 2 The expression  $\frac{3\sqrt{3} + 3}{\sqrt{3} - 1}$  can be written in the form  $a + b\sqrt{3}$ , where  $a$  and  $b$  are integers.  
What is the value of  $a + b$ ?

- A 0
- B 3
- C 6
- D 9
- E 18

- 3** A fair four-sided die has faces labelled 1, 2, 3, 4.  
A fair six-sided die has faces labelled 1, 1, 2, 3, 4, 6.

Both dice are rolled simultaneously. Let  $a$  be the score on the four-sided die and  $b$  be the score on the six-sided die.

What is the probability that the quadratic equation

$$x^2 + ax + b = 0$$

has real roots?

- A**  $\frac{7}{24}$
- B**  $\frac{3}{8}$
- C**  $\frac{5}{12}$
- D**  $\frac{11}{24}$
- E**  $\frac{7}{12}$

- 4 A square of side length  $x$  has area  $A$ .

The square is modified to form the pentagon shown in the diagram below:

- the base length  $b$  is 50% greater than  $x$ ,
- the height  $h$  is 20% less than  $x$ ,
- a right-angled triangular corner with base  $\frac{1}{3}b$  and height  $\frac{1}{2}h$  is removed from the top-right corner.

What is the area of the resulting pentagon in terms of  $A$ ?

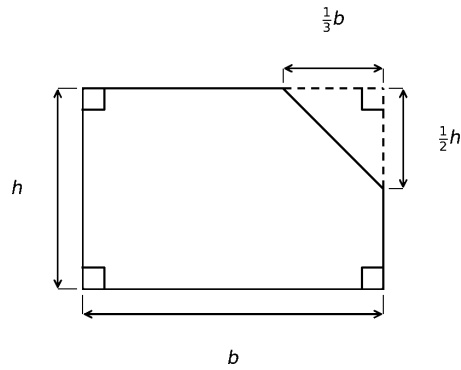


Diagram not to scale

- A  $\frac{9}{10}A$
- B  $A$
- C  $\frac{11}{10}A$
- D  $\frac{67}{60}A$
- E  $\frac{6}{5}A$
- F  $\frac{13}{10}A$

- 5 A component has a volume of  $V$  cubic inches. One inch is equivalent to  $L$  centimetres. The component is made from a material with a density of  $D \text{ g/cm}^3$ . The material is supplied in blocks, each with a mass of  $M$  kg. Each block costs  $C$  pounds.

Which of the following expressions gives the cost of the material for the component, in pence?  
(There are 100 pence in one pound and 1000 grams in one kilogram.)

- A  $\frac{VCDL^3}{10M}$   
 B  $\frac{VCDL}{10M}$   
 C  $\frac{VCD}{10ML^3}$   
 D  $\frac{100VCDL^3}{M}$   
 E  $\frac{VCDL^3}{1000M}$   
 F  $\frac{VCL^3}{10MD}$

- 6 The diagram shows a trapezium with parallel horizontal sides of length  $2^{x+1}$  and  $2^{x+3}$ , and perpendicular height  $4^{2-x}$ .

Given that the area of the trapezium is  $40\sqrt{2}$ , what is the value of  $x$ ?

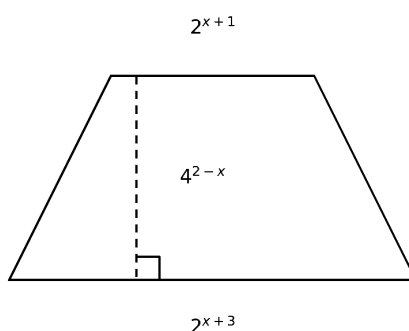


Diagram not to scale

- A  $-1.5$   
 B  $-0.5$   
 C  $0.5$   
 D  $1.5$   
 E  $2.5$

- 7 The graphs of the functions  $y = (k - 1)x^2 + 6x$  and  $y = -k - 7$  intersect at exactly one point. What is the sum of all possible real values of the constant  $k$ ?
- A -6  
B -5  
C 1  
D 2  
E 3
- 8 Two solid objects,  $X$  and  $Y$ , are mathematically similar. The volume of  $X$  is 24 and the volume of  $Y$  is 81. The total surface area of  $X$  is  $S$ . Which expression gives the total surface area of  $Y$ ?
- A  $\frac{8}{27}S$   
B  $\frac{4}{9}S$   
C  $\frac{2}{3}S$   
D  $\frac{3}{2}S$   
E  $\frac{9}{4}S$   
F  $\frac{27}{8}S$
- 9 A box contains only black discs and white discs. The total number of discs in the box is 15. There are more black discs than white discs in the box. Two discs are drawn at random from the box without replacement. The probability that both discs drawn are of the same colour is  $\frac{11}{21}$ . What is the probability that both discs drawn are black?
- A  $\frac{2}{21}$   
B  $\frac{1}{7}$   
C  $\frac{5}{21}$   
D  $\frac{2}{7}$   
E  $\frac{1}{3}$   
F  $\frac{3}{7}$   
G  $\frac{4}{9}$   
H  $\frac{2}{3}$

- 10** A set of 5 distinct integers has a mean of 10, a median of 5 and a range of 20.  
The integer 8 is added to the set.  
Which of the following correctly describes the changes to the mean, the median and the range of the set?
- A** Mean decreases, Median unchanged, Range increases
  - B** Mean increases, Median increases, Range unchanged
  - C** Mean decreases, Median increases, Range unchanged
  - D** Mean increases, Median unchanged, Range increases
  - E** Mean decreases, Median increases, Range increases
- 11** An unlimited supply of gold and silver coins is available. The probability of selecting a gold coin is  $\frac{1}{3}$ .  
Two boxes, Box 1 and Box 2, are prepared. To prepare each box, two coins are selected from the supply and placed inside.  
It is known that each box contains at least one gold coin.  
One coin is then drawn from Box 1 and one coin is drawn from Box 2.  
What is the probability that both coins drawn are gold?
- A**  $\frac{1}{25}$
  - B**  $\frac{1}{9}$
  - C**  $\frac{9}{25}$
  - D**  $\frac{3}{5}$
  - E**  $\frac{9}{16}$
  - F**  $\frac{4}{5}$

- 12** A factory processes 1200 kg of raw material in batches of 20 kg.
- In Stage 1, each batch is converted into 10 units of substance P and 5 units of substance G.
- In Stage 2, all of substance P is processed, with each unit of P producing 0.5 units of substance G.
- All the substance G from both stages is used to create a final product, F. 3 units of G are required to produce 2 units of F. The product F is produced at a constant rate of 5 units per minute.
- What is the total time, in minutes, taken to produce all of F?
- A** 40  
**B** 80  
**C** 90  
**D** 120  
**E** 180  
**F** 240
- 13** A composite cylindrical rod of length  $L$  consists of a solid inner core of radius  $R$  and a surrounding outer shell of inner radius  $R$  and outer radius  $2R$ .
- The core is made of a material with density  $\rho$  and has mass  $M$ .
- The shell is made of a material with density  $2\rho$ .
- What is the total mass of the rod?
- A**  $3M$   
**B**  $4M$   
**C**  $6M$   
**D**  $7M$   
**E**  $8M$   
**F**  $9M$
- 14** A set of  $n$  distinct items ( $n \geq 3$ ) is arranged in a random linear order. Two specific items,  $X$  and  $Y$ , are chosen from the set.
- Which expression gives the probability that item  $X$  appears earlier in the arrangement than item  $Y$ ?
- A**  $\frac{1}{n(n-1)}$   
**B**  $\frac{1}{n}$   
**C**  $\frac{2}{n}$   
**D**  $\frac{1}{2}$   
**E**  $\frac{n-1}{n}$

- 15 The quadratic equation in  $x$

$$x^2 + (4^{2-2k} \times 8^k)x + 16^{2-k} = 0$$

has exactly one real root.

What is the value of  $k$ ?

- A -3
  - B -2
  - C 0
  - D 1
  - E 2
  - F 3
- 16 Five years ago, the ratio of the age of P to the age of Q was 5 : 2. In 12 years, P will be 21 years older than Q.  
What is the current age of P?
- A 19
  - B 23
  - C 35
  - D 40
  - E 47
- 17 A 3D printer extrudes a cylindrical filament of plastic with a diameter of 4 mm. The filament is fed at a constant rate of 5 mm per second for a total of 10 minutes.  
The density of the plastic is  $\frac{2.5}{\pi}$  g/cm<sup>3</sup>.  
What is the total mass of plastic used, in grams?
- A 0.5
  - B 2
  - C 30
  - D 120
  - E 300
  - F 3000

- 18 How many integer values of  $x$  are there such that the value of the expression

$$\frac{3x + 31}{3x + 1}$$

is an integer?

- A 2
  - B 4
  - C 8
  - D 16
- 19 The line  $L_1$  passes through the point  $A(1, 2)$  and is perpendicular to the line segment connecting  $A$  and  $B(5, 4)$ . The line  $L_2$  passes through  $B$  and is perpendicular to the line  $x + 3y = 0$ . The lines  $L_1$  and  $L_2$  intersect at the point  $C$ . What is the area of the triangle  $ABC$ ?
- A 5
  - B 10
  - C 20
  - D 30
  - E 40
- 20 An equilateral triangle and a regular 12-sided polygon are inscribed in the same circle. The area of the equilateral triangle is  $27\sqrt{3}$ . What is the area of the regular 12-sided polygon?
- A 27
  - B 54
  - C 108
  - D  $108\sqrt{3}$
  - E 216
  - F 324

- 21** A cubic polynomial  $P(x)$  with leading coefficient 1 can be factorised as  $P(x) = (x - r_1)(x - r_2)(x - r_3)$ , where  $r_1, r_2$  and  $r_3$  are positive integers.

The curve  $y = P(x)$  passes through the point  $(3, -4)$ .

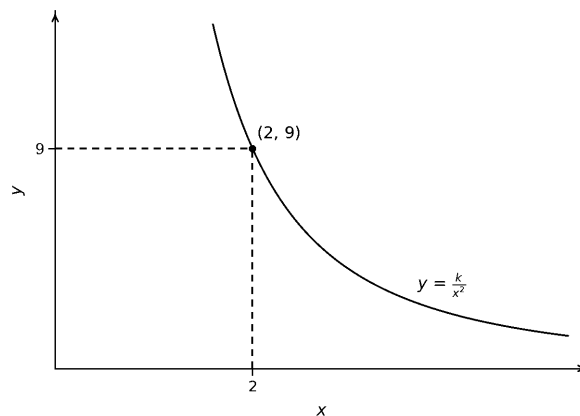
How many distinct polynomials  $P(x)$  satisfy these conditions?

- A** 1
- B** 3
- C** 4
- D** 5
- E** 6
- F** 18

- 22** The diagram shows part of the curve  $y = \frac{k}{x^2}$ , where  $k$  is a positive constant and  $x > 0$ . The curve passes through the point  $(2, 9)$ .

Also,  $x$  is directly proportional to  $z^3$ . When  $z = 2$ ,  $x = 24$ .

Which of the following is an expression for  $y$  in terms of  $z$ ?



- A**  $y = \frac{1}{4z^6}$
- B**  $y = \frac{4}{z^6}$
- C**  $y = \frac{4}{z^5}$
- D**  $y = \frac{6}{z^3}$
- E**  $y = \frac{12}{z^6}$
- F**  $y = \frac{36}{z^6}$
- G**  $y = \frac{81z^6}{4}$
- H**  $y = 4z^6$

- 23** A system's success depends on two independent paths, A and B. The system succeeds if at least one path succeeds. Each path consists of 3 relays in series. A path succeeds only if all 3 of its relays succeed. All relays are identical and independent, each with a probability  $p$  of success. In 128 independent trials, the expected number of system successes is 30.

What is the value of  $p$ ?

- A**  $\frac{1}{24}$
- B**  $\frac{1}{8}$
- C**  $\frac{3}{8}$
- D**  $\frac{1}{2}$
- E**  $\frac{5}{8}$
- F**  $\frac{7}{8}$

- 24** Three regular polygons meet at a single vertex, completely covering the angle around it. One polygon is a square. The other two have  $n$  sides and  $m$  sides respectively, where  $n < m$ . What is the maximum possible value of  $m$ ?

- A** 8
- B** 12
- C** 16
- D** 20
- E** 24
- F** 42

- 25** Three quantities  $x$ ,  $y$  and  $z$  are related by the ratios  $x : y = 3 : 2$  and  $z : y = 5 : 4$ . What is the ratio  $x : z$ ?

- A** 3 : 5
- B** 5 : 6
- C** 6 : 5
- D** 8 : 15
- E** 15 : 8

- 26** The variables  $x$  and  $y$  are positive.

For which of the following relationships is  $y$  inversely proportional to  $x$ ?

- A** The sum  $x + y$  is a constant.
- B**  $y = \frac{k}{x^2}$  for some non-zero constant  $k$ .
- C**  $y = \frac{k}{x + c}$  for some non-zero constants  $k$  and  $c$ .
- D** The product  $xy$  is a non-zero constant.
- E** The ratio  $\frac{y}{x}$  is a non-zero constant.
- F**  $y = \frac{k}{x} + c$  for some non-zero constants  $k$  and  $c$ .

- 27** The following four lengths are denoted by P, Q, R, and S.

$$P = 5\mu\text{m}$$

$$Q = 0.02\text{mm}$$

$$R = 200\text{nm}$$

$$S = 5 \times 10^{-5}\text{m}$$

Arrange these lengths in order, from smallest to largest.

- A**  $R, P, Q, S$
- B**  $S, Q, P, R$
- C**  $P, Q, S, R$
- D**  $Q, S, R, P$
- E**  $R, P, S, Q$
- F**  $R, Q, S, P$

## Mathematics 2

- 1 What is the value of the definite integral

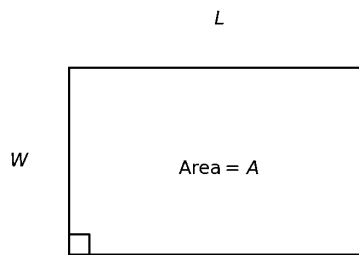
$$\int_{-2}^2 (3x^2 + 5 + x^3 \cos x) dx$$

- A 0
- B 16
- C 18
- D 36
- E 72

- 2 The diagram shows a rectangle of length  $L$ , width  $W$ , and area  $A$ .  
Given that

$$A = \frac{3x^2 - 11x - 4}{9x^2 - 1} \quad \text{and} \quad W = \frac{x^2 - 2x - 8}{3x^2 + 14x - 5}$$

which one of the following is an expression for  $L$  in its simplest form?



*Diagram not to scale*

- A  $\frac{x - 5}{x + 2}$
- B  $\frac{x - 5}{x - 2}$
- C  $\frac{x + 2}{x + 5}$
- D  $\frac{x + 5}{x - 2}$
- E  $\frac{x + 5}{x + 2}$

- 3 The function  $f(x)$  is continuous on the interval  $0 \leq x \leq 8$ . The curve  $y = f(x)$  crosses the  $x$ -axis exactly once in this interval.

Given that:

$$\int_0^8 f(x) \, dx = 2$$

and the total area of the region bounded by the curve  $y = f(x)$ , the  $x$ -axis, and the lines  $x = 0$  and  $x = 8$  is 12.

What is the area of the region lying below the  $x$ -axis?

- A 2  
B 5  
C 6  
D 7  
E 10
- 4 The table shows the number of items and the mean score for three groups of data, A, B, and C.

| Group | Number of items | Mean score |
|-------|-----------------|------------|
| A     | $n$             | 12         |
| B     | $2n$            | 18         |
| C     | $k$             | 30         |

When all three groups are combined, the overall mean score of all the items is 20.

What is the ratio  $k : n$ ?

- 5 A curve has equation  $y = f(x)$ . The derivative of the function is given by  $f'(x) = 3x^2 - 12$ . The distance between the two stationary points of the curve is  $4\sqrt{65}$ .

What is the y-intercept of the curve?

- A -16  
B 0  
C 16  
D 32  
E Cannot be determined

- 6 The diagram shows two perpendicular straight lines,  $L_1$  and  $L_2$ , in the Cartesian plane.
- Line  $L_1$  passes through the points with coordinates  $(2 - k, 3k - 1)$  and  $(k + 4, k + 5)$ , where  $k$  is a constant.
- Line  $L_2$  intersects the coordinate axes at  $(0, 4)$  and  $(6, 0)$ .
- What is the value of  $k$ ?

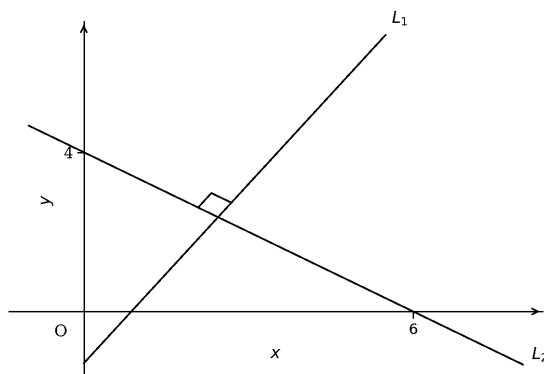


Diagram not to scale

- A  $-9$
- B  $-\frac{3}{5}$
- C  $\frac{3}{5}$
- D  $\frac{7}{5}$
- E  $\frac{5}{3}$
- F  $11$
- 7 What is the sum of all real values of  $x$  that satisfy the equation  $\sin(2x) = \cos(x)$  in the interval  $0 \leq x \leq \pi$ ?
- A  $\frac{\pi}{2}$
- B  $\frac{2\pi}{3}$
- C  $\pi$
- D  $\frac{4\pi}{3}$
- E  $\frac{3\pi}{2}$

- 8 A rhombus has diagonals of length  $2\sqrt{5}$  cm and 4 cm. An enlargement of the rhombus has an area of  $80\sqrt{5}$  cm<sup>2</sup>.

What is the perimeter of the enlarged rhombus?

- A  $6\sqrt{5}$  cm
- B  $12\sqrt{5}$  cm
- C  $24\sqrt{5}$  cm
- D  $48\sqrt{5}$  cm
- E 240 cm
- F 480 cm

- 9 In triangle  $ABC$ ,

$$\angle A = 30^\circ, \quad BC = 4, \quad CA = 4\sqrt{3}.$$

What is the sum of the areas of all possible non-congruent triangles  $ABC$  satisfying these conditions?

- A  $4\sqrt{3}$
- B  $8\sqrt{3}$
- C  $12\sqrt{3}$
- D  $16\sqrt{3}$
- E  $20\sqrt{3}$

- 10 The positive variables  $u$ ,  $v$ ,  $w$ , and  $x$  satisfy the following relationships:

- $u$  is inversely proportional to the square root of  $v$
- $v$  is directly proportional to the cube of  $w$
- $w$  is inversely proportional to the square of  $x$

Which of the following correctly describes the relationship between  $u$  and  $x$ ?

- A  $u$  is directly proportional to  $x^3$
- B  $u$  is inversely proportional to  $x^3$
- C  $u$  is directly proportional to  $x^6$
- D  $u$  is inversely proportional to  $x^6$
- E  $u$  is directly proportional to  $x^{3/2}$
- F  $u$  is inversely proportional to  $x^{3/2}$

- 11 A geometric series has a common ratio  $r$ . The value of  $r$  is the result of increasing the number 1 by  $p\%$ , and then decreasing the new value by  $p\%$ , where  $p > 0$ .

The sum to infinity of this series exists.

Which of the following gives the complete set of possible values for  $p$ ?

- A  $0 < p < 100$
  - B  $0 < p < 100\sqrt{2}$
  - C  $p > 0$
  - D  $p > 100\sqrt{2}$
  - E  $0 < p < 200$
  - F No values of  $p$ .
- 12 The expression  $(3x + 2)^6$  is expanded in ascending powers of  $x$ .  
How many of the coefficients in this expansion are divisible by 6?
- A 4
  - B 5
  - C 6
  - D 7
  - E 2
- 13 The function  $f(x) = ax^2 + bx + c$  has a stationary point at  $x = 2$ . The gradient of the curve  $y = f(x)$  at  $x = 0$  is 12. Given that  $f(1) = 15$ , what is the value of  $c$ ?
- A 0
  - B 6
  - C 9
  - D 12
  - E 24

- 14 A circle has the equation

$$x^2 + y^2 - 2x + 2y - 23 = 0$$

Find the area of the triangle bounded by the coordinate axes and the tangent to the circle at the point  $(4, 3)$ .

- A 6
- B 24
- C 27
- D 48
- E  $\frac{49}{24}$

- 15 The function  $f(x)$  is defined for  $0 \leq x \leq 6$  by:

$$f(x) = \begin{cases} 11 + 2x - x^2 & \text{for } 0 \leq x \leq 3 \\ 8x - x^2 - 6 & \text{for } 3 < x \leq 6 \end{cases}$$

What is the maximum value of  $f(x)$ ?

- A 9  
B 10  
C 11  
D 12
- 16 A function  $f$  is defined by

$$f(x) = \log_3 \left( \frac{ax + b}{x} \right)$$

where  $a$  and  $b$  are constants.

Given that  $f(1) = 2$  and  $f(4) = 1$ , find the value of  $x$  such that  $f(x) = -1$ .

- A -12  
B -6  
C -4  
D -2  
E 4
- 17 How many solutions does the equation  $2 \cos \left( 2x + \frac{\pi}{3} \right) = 1$  have in the interval  $0 \leq x \leq 2\pi$ ?
- A 2  
B 3  
C 4  
D 5  
E 6
- 18 Quantities  $X$  and  $Y$  change in discrete steps  $n = 0, 1, 2, \dots$ . Initially,  $X_0 = 20Y_0$ , where  $Y_0 > 0$ . In each step,  $X$  decreases by 60% of its value at the start of the step, and  $Y$  decreases by 20% of its value at the start of the step. What is the minimum value of  $n$  such that  $X_n < Y_n$ ?
- A 2  
B 4  
C 5  
D 6

- 19** The curve with equation  $y = 4^x - 10 \cdot 2^x$  intersects the horizontal line  $y = c$  at two distinct points. The  $x$ -coordinates of these points are  $p$  and  $q$ , where  $q > p$ . Given that  $q - p = 2$ , what is the value of the constant  $c$ ?
- A**  $-25$   
**B**  $-24$   
**C**  $-16$   
**D**  $-4$   
**E**  $16$
- 20** A circle passes through the points  $(2, 1)$  and  $(4, 5)$  and is tangent to the  $x$ -axis. What is the largest possible radius of the circle?
- A**  $1$   
**B**  $2.5$   
**C**  $3$   
**D**  $5$   
**E**  $5\sqrt{2}$
- 21** Let  $n$  be a positive odd integer. Two distinct coefficients in the expansion of  $(1 + x)^n$  are chosen at random. The probability that these two coefficients are equal is  $\frac{1}{23}$ . What is the value of  $n$ ?
- A**  $21$   
**B**  $22$   
**C**  $23$   
**D**  $24$   
**E**  $25$   
**F**  $45$
- 22** A sequence of three positive integers has a sum of 6 and a sum of the squares of its terms of 18. How many such distinct sequences are there?
- A**  $0$   
**B**  $1$   
**C**  $2$   
**D**  $3$   
**E**  $6$   
**F**  $9$

- 23** The graph of  $y = f(x)$  has exactly three  $x$ -intercepts:

$$-4, \quad 2, \quad 8.$$

A new function is defined by

$$g(x) = 3f(2x - 4).$$

What is the sum of the  $x$ -coordinates of the  $x$ -intercepts of  $y = g(x)$ ?

- A** 1  
**B** 3  
**C** 5  
**D** 7  
**E** 9  
**F** 11
- 24** A particle travels at a constant speed  $v$  from the point  $(-2R, 0)$  to the point  $(2R, 0)$ , where  $R > 0$ . The particle's path is chosen to be the shortest possible route that does not enter the region  $x^2 + y^2 < R^2$ .

What is the minimum time for this journey?

- A**  $\frac{R}{v} \left( 2\sqrt{3} + \frac{\pi}{3} \right)$   
**B**  $\frac{R}{v} (\pi + 2)$   
**C**  $\frac{R}{v} \left( 2\sqrt{3} + \frac{2\pi}{3} \right)$   
**D**  $\frac{R}{v} (2\sqrt{5})$   
**E**  $\frac{R}{v} (2\sqrt{3})$

- 25** A function  $f$  is defined by

$$f(x) = \frac{x^2 + 5x + 4}{x^2 + 7x + 12} \div \frac{x + 1}{3x + 9}$$

for all values of  $x$  for which the expression is defined.

What is the value of  $f'(4)$ ?

- A** 0  
**B**  $\frac{1}{3}$   
**C** 1  
**D** 3  
**E** 4  
**F** -3

- 26** A geometric progression has first term  $a$  and common ratio  $r$ .  
The first term is

$$a = \frac{1}{x+2}$$

and the sum to infinity is

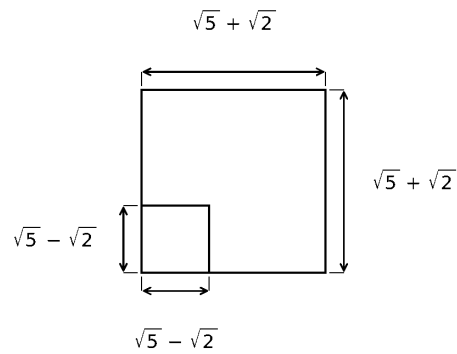
$$S_{\infty} = \frac{x-1}{x^2-4}$$

What is the common ratio  $r$ ?

- A**  $\frac{1}{x-1}$
- B**  $\frac{x-2}{x-1}$
- C**  $\frac{1}{2-x}$
- D**  $\frac{2x-3}{x-1}$
- E**  $-\frac{1}{x-1}$
- F**  $\frac{x+1}{x+2}$

- 27 The diagram shows a square of side length  $\sqrt{5} + \sqrt{2}$ . A smaller square of side length  $\sqrt{5} - \sqrt{2}$  is removed from one corner, leaving the shaded region.

What is the area of the shaded region?



*Diagram not to scale*

- A 4
- B 8
- C 14
- D  $2\sqrt{10}$
- E  $4\sqrt{10}$
- F  $7 + 2\sqrt{10}$
- G  $14 - 4\sqrt{10}$
- H  $14 + 4\sqrt{10}$

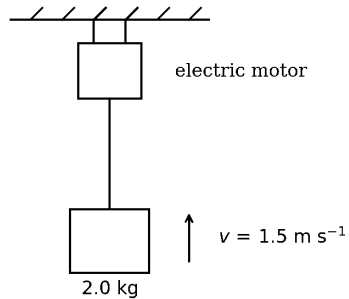
## Physics

- 1 An electric motor is used to lift a block of mass  $2.0 \text{ kg}$  vertically at a constant speed of  $1.5 \text{ m s}^{-1}$ , as shown in the diagram.

The electrical power supplied to the motor is  $50 \text{ W}$ .

How much energy is wasted by the motor in  $5.0$  minutes?

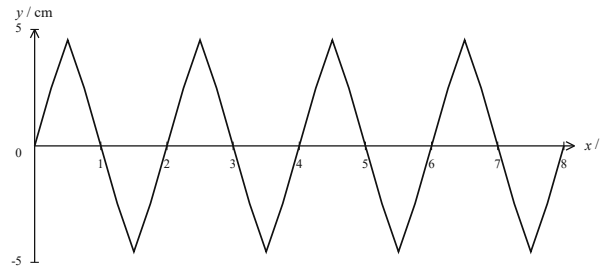
[gravitational field strength =  $10 \text{ N kg}^{-1}$ ]



*Diagram not to scale*

- A**  $100 \text{ J}$   
**B**  $150 \text{ J}$   
**C**  $6000 \text{ J}$   
**D**  $9000 \text{ J}$   
**E**  $15\,000 \text{ J}$
- 2 A stationary neutral atom contains a total of  $N$  constituent particles (protons, neutrons and electrons).  
 The atom undergoes alpha decay, and the resulting daughter product is also a neutral atom.  
 What is the total number of constituent particles in the daughter atom?
- A**  $N - 8$   
**B**  $N - 6$   
**C**  $N - 4$   
**D**  $N - 2$   
**E**  $N$   
**F**  $N + 1$

- 3 The graph shows a snapshot of a transverse wave on a string, with displacement  $y$  in cm and position  $x$  in m. The wave travels at a speed of  $10 \text{ m s}^{-1}$ .



What is the average speed of a particle on the string during one complete oscillation?

- A  $0.50 \text{ m s}^{-1}$
  - B  $1.0 \text{ m s}^{-1}$
  - C  $2.0 \text{ m s}^{-1}$
  - D  $4.0 \text{ m s}^{-1}$
  - E  $10 \text{ m s}^{-1}$
  - F  $\pi \text{ m s}^{-1}$
- 4 A vehicle of constant mass  $m$  accelerates from rest along a horizontal straight track. The engine provides a constant mechanical power output  $P$ , and all resistive forces are negligible.
- What is happening to the magnitude of the acceleration of the vehicle as time increases?
- A It is increasing at an increasing rate.
  - B It is increasing at a constant rate.
  - C It is increasing at a decreasing rate.
  - D It is not changing.
  - E It is decreasing at an increasing rate.
  - F It is decreasing at a constant rate.
  - G It is decreasing at a decreasing rate.

- 5 In a mass spectrometer, the radius of a nucleus's circular path is proportional to its mass-to-charge ratio. An unknown nucleus is found to have a mass-to-charge ratio that is 2.5 times that of a proton.

Assuming a nucleus's mass is proportional to its mass number  $A$  and its charge is proportional to its atomic number  $Z$ , which of the following could be the unknown nucleus? The nuclei are described by (number of protons, number of neutrons).

- A (2, 2)
- B (2, 3)
- C (2, 4)
- D (2, 5)
- E (3, 2)
- F (5, 2)

- 6 An ultrasound pulse enters a layer of tissue along a line perpendicular to its two parallel surfaces. Echoes from the front and back surfaces return to the transmitter  $30\ \mu\text{s}$  and  $70\ \mu\text{s}$  after emission.

The speed of ultrasound in the layer is  $1500\ \text{m s}^{-1}$ . What is the thickness of the layer?

- A 1.5 cm
- B 3.0 cm
- C 4.5 cm
- D 6.0 cm
- E 7.5 cm

- 7 Narrow beams of alpha particles, beta-minus particles and gamma radiation enter the region between two charged plates. The beams initially travel horizontally to the right. The upper plate is positive and the lower plate is negative.

Which row gives the direction in which each beam bends?

- A alpha downwards; beta upwards; gamma straight
- B alpha upwards; beta downwards; gamma straight
- C alpha downwards; beta straight; gamma upwards
- D alpha upwards; beta upwards; gamma straight
- E alpha straight; beta upwards; gamma downwards

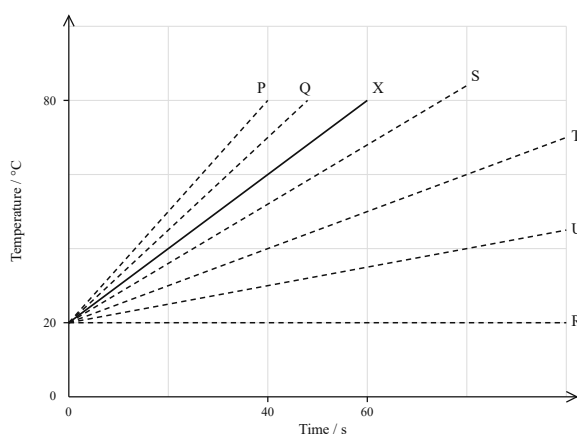
- 8 A rock sample is known to have initially consisted entirely of a radioactive parent isotope P. This isotope decays to a single stable daughter isotope D. At a time  $t$ , the ratio of the number of atoms of D to the number of atoms of P is 3 : 1.

What is the ratio of the number of atoms of D to the number of atoms of P at time  $2t$ ?

- A 6 : 1
- B 7 : 1
- C 9 : 1
- D 15 : 1
- E 31 : 1
- F 63 : 1

- 9 An electric heating element consists of three identical resistors connected in series to a power supply of constant voltage. The element is used to heat a fixed mass of water, starting from an initial temperature of 20 °C.

The solid line, Graph X, in the diagram shows the temperature of the water as a function of time. No heat is lost to the surroundings.



One of the resistors fails and becomes a short circuit. The experiment is repeated with the same mass of water from the same initial temperature.

Which of the dashed lines (P, Q, R, S, T, or U) represents the new temperature–time graph?

- A P
- B Q
- C R
- D S
- E T
- F U

- 10** Two electric heaters, with power ratings of 150 W and 100 W, are used in an experiment. Each heater is used to raise the temperature of an identical mass of a liquid by the same temperature difference.

The 150 W heater completes this process 60 s faster than the 100 W heater.

Assuming no energy is lost to the surroundings, what is the total thermal energy transferred to the liquid?

- A** 3.0 kJ
- B** 3.6 kJ
- C** 6.0 kJ
- D** 9.0 kJ
- E** 18 kJ

- 11** Two long straight wires P and Q carry equal currents out of the page. Point M is midway between them. The magnetic field at M due to either wire alone has magnitude  $B$ . Other magnetic fields are negligible.

The current in Q is reversed without changing its magnitude. What is the resultant magnetic field at M?

- A** zero
- B**  $B$ , upwards on the page
- C**  $B$ , downwards on the page
- D**  $2B$ , downwards on the page
- E**  $2B$ , upwards on the page

- 12** A rigid square frame  $PQRS$ , made from uniform conducting wire of constant cross-section and resistance per unit length, lies in the plane of the page.

A uniform magnetic field is directed vertically upwards (towards the top of the page), as shown in the diagram.

A steady current  $I$  enters the frame at corner  $P$  and leaves the frame at the diagonally opposite corner  $R$ .

Which statement correctly describes the initial motion of the frame immediately after the current is applied?

(Assume the frame is in free space and unaffected by gravity or external supporting forces.)

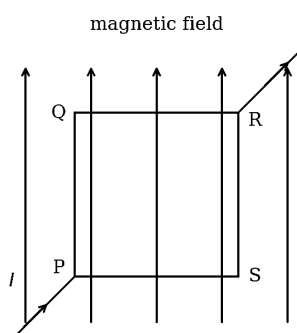


Diagram not to scale

- A** The frame moves perpendicularly out of the plane of the page without rotating.
  - B** The frame moves perpendicularly into the plane of the page without rotating.
  - C** The frame moves towards the right of the page.
  - D** The frame moves towards the top of the page.
  - E** The frame rotates about a horizontal axis in the plane of the page.
  - F** The frame rotates about a vertical axis in the plane of the page.
  - G** The frame rotates about an axis perpendicular to the plane of the page in a clockwise direction.
  - H** The frame remains completely stationary.
- 13** A heater supplies power at a constant rate of 50 W to a 2.0 kg block of material. The specific heat capacity of the material is  $500 \text{ J kg}^{-1} \text{ K}^{-1}$ .
- Over a period of 200 s, the temperature of the block increases by 8.0 K.
- What is the average rate of energy loss from the block to the surroundings during this period?
- A** 10 W
  - B** 30 W
  - C** 40 W
  - D** 50 W
  - E** 90 W

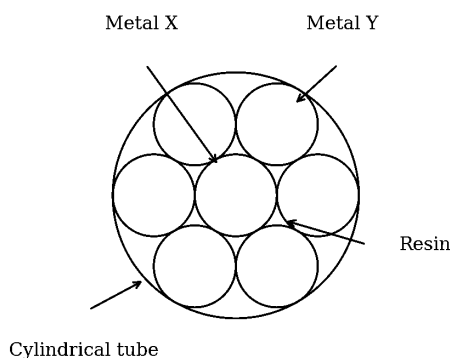
- 14** The north pole of a permanent magnet is brought close to end X of an initially unmagnetised soft-iron bar. The far end of the bar is Y.

A small compass is placed just beyond Y, on the axis of the bar. Which way does the north-seeking end of the compass point, and why is soft iron suitable for an electromagnet that must release its load when switched off?

- A** towards Y; it is difficult to magnetise
  - B** towards Y; it retains a strong magnetisation
  - C** away from Y; it retains a strong magnetisation
  - D** towards Y; it loses most of its magnetisation when the applied field is removed
  - E** away from Y; it loses most of its magnetisation when the applied field is removed
- 15** A bundle of seven identical solid cylindrical rods, each of radius 1.0 cm and length 10.0 cm, is packed symmetrically inside a hollow cylindrical tube of internal radius 3.0 cm and length 10.0 cm, as shown in the cross-section below.

The central rod is made of metal X with density  $8.0 \text{ g cm}^{-3}$ , and the six outer rods are made of metal Y with density  $3.0 \text{ g cm}^{-3}$ . The remaining space inside the tube is completely filled with a resin of density  $1.5 \text{ g cm}^{-3}$ .

What is the total mass of the contents inside the tube?



*Diagram not to scale*

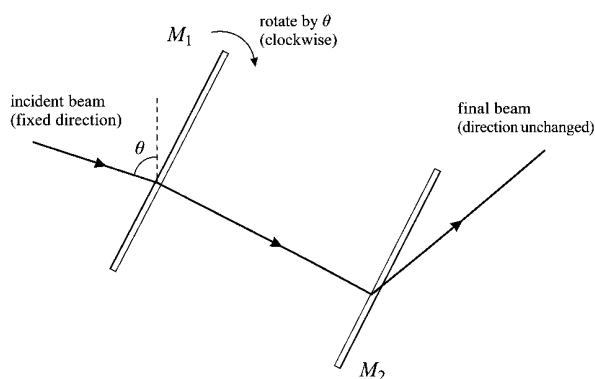
- A**  $240\pi \text{ g}$
- B**  $260\pi \text{ g}$
- C**  $290\pi \text{ g}$
- D**  $305\pi \text{ g}$
- E**  $340\pi \text{ g}$
- F**  $540\pi \text{ g}$

- 16** A stationary radioactive nucleus decays, emitting an alpha particle of mass  $m_\alpha$  and leaving a daughter nucleus of mass  $m_d$ . The kinetic energy of the emitted alpha particle is  $K$ .

What is the kinetic energy of the recoiling daughter nucleus?

(Assume non-relativistic speeds.)

- A** 0
- B**  $K$
- C**  $K \frac{m_d}{m_\alpha}$
- D**  $K \frac{m_\alpha}{m_d}$
- E**  $K \frac{m_\alpha}{m_d + m_\alpha}$
- F**  $K \frac{m_d}{m_d + m_\alpha}$
- 17** A laser beam reflects from a plane mirror  $M_1$  onto a second plane mirror  $M_2$ , and then emerges in a final direction. The path of the beam lies within a 2D plane. Initially, both mirrors are stationary.



$M_1$  is then rotated by an angle  $\theta$  in a clockwise direction. The incident beam direction remains fixed.

To keep the final direction of the beam emerging from  $M_2$  unchanged, what rotation must be applied to  $M_2$ ?

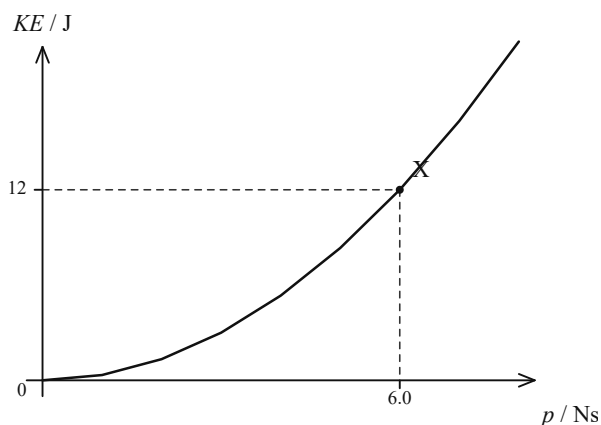
- A**  $\theta$  clockwise
- B**  $\theta$  anticlockwise
- C**  $2\theta$  clockwise
- D**  $2\theta$  anticlockwise

- 18** A block with a total volume of  $1.0 \text{ m}^3$  is constructed from two materials, A and B. The density of material A is  $200 \text{ kg m}^{-3}$  and the density of material B is  $1200 \text{ kg m}^{-3}$ .

The block floats in water with exactly  $\frac{4}{5}$  of its volume submerged. The density of water is  $1000 \text{ kg m}^{-3}$ .

What is the mass of material B in the block?

- A** 80 kg
  - B** 600 kg
  - C** 720 kg
  - D** 800 kg
  - E** 960 kg
- 19** The graph shows the kinetic energy, KE, of an object plotted against the magnitude of its momentum,  $p$ . A point X on the graph corresponds to a momentum of  $6.0 \text{ N s}$  and a kinetic energy of  $12 \text{ J}$ .



What is the speed of the object at this point?

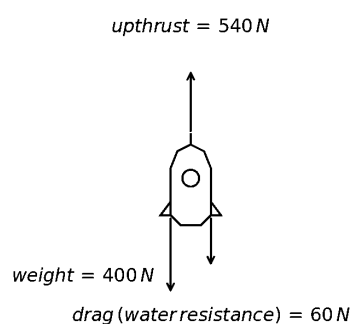
- A**  $0.50 \text{ m s}^{-1}$
- B**  $1.5 \text{ m s}^{-1}$
- C**  $2.0 \text{ m s}^{-1}$
- D**  $4.0 \text{ m s}^{-1}$
- E**  $6.0 \text{ m s}^{-1}$
- F**  $9.0 \text{ m s}^{-1}$

- 20** An ideal transformer supplies a fixed resistor. The primary voltage is held constant. The transformer is replaced with another ideal transformer having half as many primary turns and the same number of secondary turns. The same resistor is connected across the secondary. By what factor does the primary current change?

- A**  $\frac{1}{4}$   
**B**  $\frac{1}{2}$   
**C** 1  
**D** 2  
**E** 4

- 21** An autonomous underwater probe of mass 40 kg is ascending vertically through water. At a particular instant, the vertical forces acting on the probe are shown in the diagram. Which row in the table gives the magnitude and direction of the probe's acceleration at this instant?

|   | magnitude of acceleration / $\text{m s}^{-2}$ | direction of acceleration |
|---|-----------------------------------------------|---------------------------|
| A | 2.0                                           | upwards                   |
| B | 2.0                                           | downwards                 |
| C | 3.5                                           | upwards                   |
| D | 3.5                                           | downwards                 |
| E | 5.0                                           | upwards                   |
| F | 12.0                                          | upwards                   |



[diagram not to scale]

- 22** A shallow dish of water is left in a room at constant temperature. For a short time, evaporation cools the remaining water below room temperature.

Which pair of statements explains this cooling?

- A** Faster particles escape preferentially; the average kinetic energy of the remaining particles decreases.
  - B** Slower particles escape preferentially; the average kinetic energy of the remaining particles decreases.
  - C** Faster particles escape preferentially; the remaining particles become smaller.
  - D** All particles slow down by the same amount before any can escape.
  - E** Evaporation can occur only after every particle has the same kinetic energy.
- 23** A detector is used to monitor the radiation emitted by a sample of a single radioactive isotope. The detector records the total count rate, which includes a constant background count rate.
- The recorded count rate at three different times is:
- at  $t = 0$  minutes, count rate = 420 counts per minute
  - at  $t = 40$  minutes, count rate = 120 counts per minute
  - at  $t = 80$  minutes, count rate = 45 counts per minute

Which row in the table gives the half-life of the isotope and the background count rate?

|          | Half-life / min | Background count rate / counts per min |
|----------|-----------------|----------------------------------------|
| <b>A</b> | 20              | 15                                     |
| <b>B</b> | 20              | 20                                     |
| <b>C</b> | 20              | 25                                     |
| <b>D</b> | 40              | 15                                     |
| <b>E</b> | 40              | 20                                     |
| <b>F</b> | 40              | 25                                     |
| <b>G</b> | 80              | 20                                     |
| <b>H</b> | 80              | 45                                     |

- 24** A bat flies directly towards a wall at a constant speed of  $5.0 \text{ m s}^{-1}$ . When it is a distance  $d$  from the wall, it emits a short ultrasound pulse. The echo is received  $0.120 \text{ s}$  later. The speed of sound is  $340 \text{ m s}^{-1}$ .

What was the distance  $d$  when the pulse was emitted?

- A** 20.1 m
  - B** 20.4 m
  - C** 20.7 m
  - D** 21.0 m
  - E** 40.8 m
  - F** 41.4 m
- 25** A singly positively charged ion  $\text{P}^+$  of a particular element contains a total of 23 particles (protons, neutrons, and electrons).

The table shows information about the total number of particles and the relative charges of four atoms or ions W, X, Y, and Z.

| atom or ion | total number of particles | relative charge |
|-------------|---------------------------|-----------------|
| W           | 24                        | 0               |
| X           | 25                        | 0               |
| Y           | 26                        | -1              |
| Z           | 22                        | +1              |

Which of these atoms or ions could be of a different isotope to  $\text{P}^+$  but of the same element as  $\text{P}^+$ ?

- A** X only
- B** Y only
- C** W and Z only
- D** X and Y only
- E** X and Z only
- F** W, X and Y only
- G** W, Y and Z only
- H** X, Y and Z only

- 26** A wave travels 8.4 m in 0.70 s. The source produces 2.0 complete oscillations each second. What is the wavelength of the wave?
- A** 1.4 m
  - B** 2.4 m
  - C** 4.2 m
  - D** 6.0 m
  - E** 8.4 m
  - F** 12.0 m
- 27** A continuous progressive wave travels from Medium 1 into Medium 2. The speed of the wave decreases by 25% when it enters Medium 2. If the wavelength of the wave in Medium 1 is  $\lambda$ , what is the wavelength of the wave in Medium 2?
- A**  $\frac{\lambda}{2}$
  - B**  $\frac{2\lambda}{3}$
  - C**  $\frac{3\lambda}{4}$
  - D**  $\lambda$
  - E**  $\frac{4\lambda}{3}$

## Chemistry

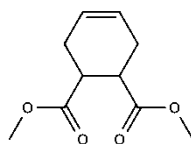
- 1 A sample of element X consists of only three isotopes:  $^{68}\text{X}$ ,  $^{70}\text{X}$ , and  $^{72}\text{X}$ . The relative atomic mass ( $A_r$ ) of X in this sample is 70.2.

| isotope         | percentage abundance / % |
|-----------------|--------------------------|
| $^{68}\text{X}$ | $x$                      |
| $^{70}\text{X}$ | 30                       |
| $^{72}\text{X}$ | $y$                      |

What is the value of  $y$ ?

- A** 20  
**B** 30  
**C** 35  
**D** 40  
**E** 45
- 2 Magnesium is an element in Group 2 of the Periodic Table.  
 Which of the following statements is/are correct about magnesium?
- 1 Magnesium is a stronger reducing agent than calcium.  
 2 Magnesium has a smaller atomic radius than calcium.  
 3 Magnesium reacts with nitrogen gas to form a compound containing 72% magnesium by mass.  
 ( $A_r$  values: N = 14; Mg = 24; Ca = 40)
- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

- 3 The skeletal formula of an organic compound is shown below.  
What is the empirical formula of this compound?



- A  $C_5H_7O_2$   
 B  $C_{10}H_{14}O_4$   
 C  $C_5H_6O_2$   
 D  $C_5H_8O_2$   
 E  $C_4H_7O_2$   
 F  $C_2H_3O$
- 4 After water vapour and carbon dioxide have been removed, liquefied air contains mainly nitrogen, argon and oxygen. Their boiling points are shown.

As the liquid is warmed slowly, in which order are the gases collected first to last?

| gas      | boiling point / $^{\circ}\text{C}$ |
|----------|------------------------------------|
| nitrogen | $-196^{\circ}\text{C}$             |
| argon    | $-186^{\circ}\text{C}$             |
| oxygen   | $-183^{\circ}\text{C}$             |

- A oxygen, argon, nitrogen  
 B nitrogen, oxygen, argon  
 C nitrogen, argon, oxygen  
 D argon, nitrogen, oxygen  
 E argon, oxygen, nitrogen

- 5** Two experiments were carried out using separate samples of an aqueous solution of an unknown base B of concentration  $0.150 \text{ mol dm}^{-3}$  ( $11.1 \text{ g dm}^{-3}$ ), and the following observations were made:

\*  $20.0 \text{ cm}^3$  of the base solution was completely neutralised by  $30.0 \text{ cm}^3$  of  $0.200 \text{ mol dm}^{-3}$  hydrochloric acid,  $\text{HCl(aq)}$ .

\* A pale blue precipitate formed when a few drops of aqueous copper(II) sulfate,  $\text{CuSO}_4(\text{aq})$ , were added to a portion of the base solution.

Which of the following statements can be deduced from this information?

1. One mole of the base reacts with two moles of  $\text{H}^+$  ions in complete neutralisation.
2. The base is fully dissociated in aqueous solution.
3. The relative formula mass ( $M_r$ ) of the base is 74.

( $A_r$  values are not required)

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3
- 6** What are the oxidation states of sulfur in  $\text{SO}_2$  and  $\text{SO}_4^{2-}$ , respectively?
- A** +2 and +4
- B** +4 and +6
- C** +4 and +4
- D** +6 and +4
- E** -4 and -6
- F** 0 and +6

- 7 Which particle diagram best represents a liquid? The particles are shown at one instant and are not drawn to scale.

- A A
- B B
- C C
- D D

- 8 Pollutant P is formed when a carbon-containing fuel burns with insufficient oxygen. Pollutant Q can form inside a hot car engine when nitrogen and oxygen from the air react.

Which pair correctly identifies P and Q?

- A P: carbon monoxide; Q: nitrogen oxides
- B P: carbon dioxide; Q: sulfur dioxide
- C P: sulfur dioxide; Q: carbon monoxide
- D P: nitrogen oxides; Q: carbon dioxide
- E P: methane; Q: carbon monoxide

- 9 A diol reacts with an excess of propanoic acid in the presence of an acid catalyst to form a diester.

The diester has a relative molecular mass ( $M_r$ ) of 174.

What is the relative molecular mass ( $M_r$ ) of the diol?

( $A_r$  values: H = 1.0; C = 12.0; O = 16.0)

- A 26
- B 44
- C 62
- D 76
- E 90
- F 100
- G 118
- H 136

- 10** Researchers intend to isolate mitochondria from a liver cell homogenate using differential centrifugation. The table below displays the minimum gravitational force (g) required to pellet specific cellular components.

| Component    | Minimum Force to Pellet (g) |
|--------------|-----------------------------|
| Nuclei       | 1,000                       |
| Mitochondria | 15,000                      |
| Ribosomes    | 100,000                     |

The homogenate is first centrifuged at 1,000 g for 10 minutes, producing a pellet and a supernatant.

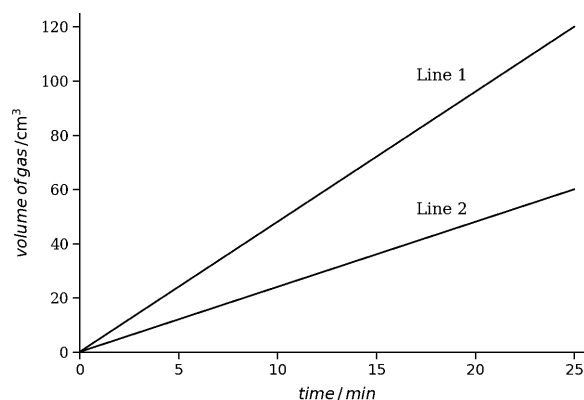
Which of the following subsequent steps would yield the purest sample of mitochondria?

- A** Resuspend the pellet from the first step and centrifuge at 15 000 *g*.
  - B** Centrifuge the supernatant from the first step at 15 000 *g* and collect the resulting pellet.
  - C** Centrifuge the supernatant from the first step at 15 000 *g* and collect the resulting supernatant.
  - D** Centrifuge the supernatant from the first step at 100 000 *g* and collect the resulting pellet.
  - E** Centrifuge the supernatant from the first step at 100 000 *g* and collect the resulting supernatant.
- 11** An aqueous solution gives a lilac flame. After acidification with dilute nitric acid, aqueous silver nitrate gives a cream precipitate. Which compound could be present?
- A** LiCl
  - B** NaBr
  - C** KCl
  - D** KBr
  - E**  $\text{CaI}_2$
  - F**  $\text{CuBr}_2$

- 12 A pure metal and an alloy made from it have the properties shown. Which statement is best supported by the data?

| material   | tensile strength / MPa | percentage extension before breaking |
|------------|------------------------|--------------------------------------|
| pure metal | 210                    | 42                                   |
| alloy      | 510                    | 13                                   |

- A The alloy is weaker and more ductile than the pure metal.  
 B The alloy is stronger and less ductile than the pure metal.  
 C The alloy must conduct electricity better than the pure metal.  
 D The alloy must have a lower density than the pure metal.  
 E The two materials have identical mechanical properties.
- 13 An aqueous solution of dilute sulfuric acid,  $\text{H}_2\text{SO}_4(\text{aq})$ , is electrolysed at room temperature and pressure using inert platinum electrodes and a constant electric current.
- The graph shows the volume of gas collected at each of the two electrodes as a function of time.
- Which of the following statements is/are correct?
- Line 1 represents the gas collected at the cathode.
  - In the external circuit, electrons flow from the electrode producing the gas of Line 1 towards the electrode producing the gas of Line 2.
  - Doubling the electric current would double the gradient of both Line 1 and Line 2.



- A none of them  
 B 1 only  
 C 2 only  
 D 3 only  
 E 1 and 2 only  
 F 1 and 3 only  
 G 2 and 3 only  
 H 1, 2 and 3

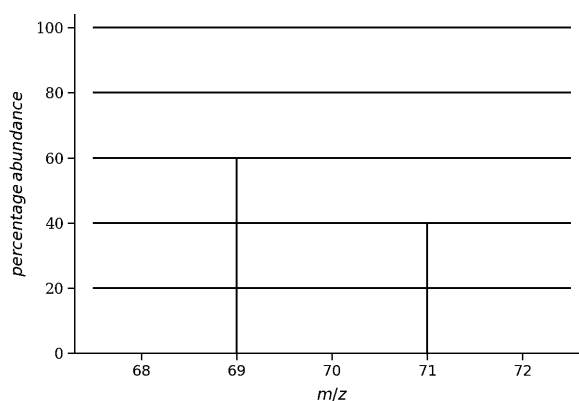
- 14 The mass spectrum of a sample of element M is shown in the diagram.

A 0.100 mol sample of M reacts completely with 2.40 g of oxygen gas ( $O_2$ ) to form a single oxide.

Which row in the table gives the relative atomic mass of this sample of M and the empirical formula of the oxide formed?

( $A_r(O) = 16.0$ )

|   | Relative atomic mass | Empirical formula of oxide |
|---|----------------------|----------------------------|
| A | 69.8                 | MO                         |
| B | 69.8                 | $M_2O_3$                   |
| C | 69.8                 | $MO_2$                     |
| D | 70.0                 | $M_2O_3$                   |
| E | 70.0                 | $MO_2$                     |
| F | 70.2                 | MO                         |
| G | 70.2                 | $M_2O_3$                   |
| H | 70.2                 | $MO_2$                     |



**15** Which of the following statements is/are correct?

1. Ethene decolourises bromine water.
2. Addition polymerisation of ethene produces a small-molecule by-product.
3. Ethanol reacts with sodium to release hydrogen gas.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

**16** A solution contains both hydrochloric acid,  $\text{HCl(aq)}$ , and sulfuric acid,  $\text{H}_2\text{SO}_4\text{(aq)}$ .

- A  $50.0 \text{ cm}^3$  sample of this solution is completely neutralized by  $35.0 \text{ cm}^3$  of  $0.200 \text{ mol dm}^{-3} \text{ NaOH(aq)}$ .
- A separate  $50.0 \text{ cm}^3$  sample of the same solution is treated with excess aqueous barium chloride,  $\text{BaCl}_2\text{(aq)}$ , producing  $0.466 \text{ g}$  of dry  $\text{BaSO}_4\text{(s)}$  precipitate.

What is the concentration of hydrochloric acid in the original solution?

( $A_r$  values:  $\text{H} = 1.0$ ;  $\text{O} = 16.0$ ;  $\text{S} = 32.0$ ;  $\text{Ba} = 137.0$ )

- A**  $0.020 \text{ mol dm}^{-3}$
- B**  $0.030 \text{ mol dm}^{-3}$
- C**  $0.040 \text{ mol dm}^{-3}$
- D**  $0.060 \text{ mol dm}^{-3}$
- E**  $0.080 \text{ mol dm}^{-3}$
- F**  $0.100 \text{ mol dm}^{-3}$

- 17 Three separate experiments are carried out. In each experiment, an excess of dilute hydrochloric acid is reacted with 0.050 mol of a solid Group 1 carbonate:

- Experiment 1:  $\text{Li}_2\text{CO}_3$
- Experiment 2:  $\text{Na}_2\text{CO}_3$
- Experiment 3:  $\text{K}_2\text{CO}_3$

Let  $V_1$ ,  $V_2$ , and  $V_3$  be the final volumes of gas produced in Experiments 1, 2, and 3 respectively, measured at room temperature and pressure.

Which of the following correctly compares these volumes?

- A  $V_1 < V_2 < V_3$
  - B  $V_1 > V_2 > V_3$
  - C  $V_1 = V_2 = V_3$
  - D  $V_1 = V_2 < V_3$
  - E  $V_1 > V_2 = V_3$
- 18 The ions  $\text{X}^{2-}$ ,  $\text{Y}^-$  and  $\text{Z}^{2+}$  are isoelectronic (they have the same electronic configuration). Which of the following statements about the **neutral atoms** X, Y and Z is correct?

- A Element Z has the lowest atomic number.
- B Element Z is in the period immediately following elements X and Y.
- C Elements X, Y and Z are in the same period.
- D Element X is in Group 2 of the Periodic Table.
- E Element Y has a larger atomic radius than element Z.

- 19 Electrolysis is carried out on separate samples of the following four aqueous solutions using inert graphite electrodes:

1.  $\text{AgNO}_3(\text{aq})$  2.  $\text{Ca}(\text{NO}_3)_2(\text{aq})$  3.  $\text{CuSO}_4(\text{aq})$  4.  $\text{NaCl}(\text{aq})$

In which of these experiments is the metal deposited at the cathode?

- A 1 and 3 only
- B 2 and 4 only
- C 1, 2 and 3 only
- D 3 and 4 only
- E 1, 2, 3 and 4

- 20** The boiling points of the following three compounds are  $-100^{\circ}\text{C}$ ,  $-24^{\circ}\text{C}$ , and  $13^{\circ}\text{C}$  (not necessarily in that order): 1. Boron trifluoride ( $\text{BF}_3$ ) 2. Chloromethane ( $\text{CH}_3\text{Cl}$ ) 3. Boron trichloride ( $\text{BCl}_3$ )

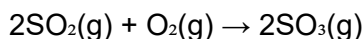
Which option lists the compounds in order of **increasing** boiling point? (Ar values:  $\text{H} = 1$ ;  $\text{B} = 11$ ;  $\text{C} = 12$ ;  $\text{F} = 19$ ;  $\text{Cl} = 35.5$ )

- A**  $\text{BF}_3 < \text{CH}_3\text{Cl} < \text{BCl}_3$
  - B**  $\text{BF}_3 < \text{BCl}_3 < \text{CH}_3\text{Cl}$
  - C**  $\text{CH}_3\text{Cl} < \text{BF}_3 < \text{BCl}_3$
  - D**  $\text{BCl}_3 < \text{CH}_3\text{Cl} < \text{BF}_3$
- 21** Compound P has condensed structural formula  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ . Compound Q has condensed structural formula  $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$ .

Which statement is correct?

- A** P and Q have different molecular formulae.
- B** P and Q are structural isomers.
- C** P is an alkane but Q is an alkene.
- D** P and Q differ only in the positions of their hydrogen atoms in space.
- E** P and Q must have identical physical properties.

- 22** The reaction energy profile for the oxidation of sulfur dioxide is shown below:

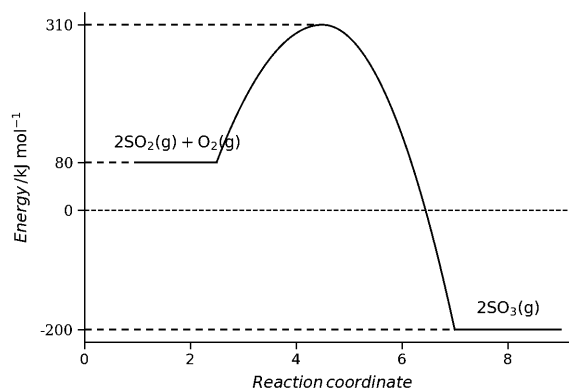


In an experiment, the thermal energy released by the complete oxidation of a sample of  $\text{SO}_2(\text{g})$  is transferred to 600 g of water. The heat transfer process is 75% efficient.

The temperature of the water increases by  $25^\circ\text{C}$ .

What mass of  $\text{SO}_2$  was reacted?

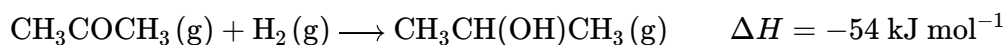
( $A_r$  values: S = 32.0, O = 16.0. Specific heat capacity of water =  $4.2 \text{ J g}^{-1}\text{C}^{-1}$ )



- A** 10.8 g
  - B** 19.2 g
  - C** 21.6 g
  - D** 38.4 g
  - E** 48.0 g
  - F** 53.8 g
- 23** A dry mixture contains sand and copper(II) sulfate crystals. Which sequence would allow a student to obtain both the sand and pure copper(II) sulfate crystals?
- A** Add water, filter, then crystallise the filtrate.
  - B** Heat strongly, then use a separating funnel.
  - C** Add water, evaporate the whole mixture to dryness, then filter.
  - D** Use fractional distillation, then centrifuge the residue.
  - E** Filter the dry mixture, then add water to the residue.

- 24 The structure of propan-2-ol is shown below:

Propanone reacts with hydrogen gas to form propan-2-ol according to the following equation:



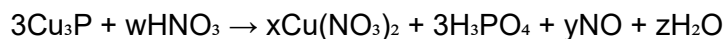
Using the mean bond enthalpies provided below, what is the mean bond enthalpy of the C=O bond in propanone?

Mean bond enthalpies:

- C-C:  $348 \text{ kJ mol}^{-1}$
- C-H:  $413 \text{ kJ mol}^{-1}$
- C-O:  $358 \text{ kJ mol}^{-1}$
- H-H:  $436 \text{ kJ mol}^{-1}$
- O-H:  $464 \text{ kJ mol}^{-1}$



- A  $332 \text{ kJ mol}^{-1}$
  - B  $440 \text{ kJ mol}^{-1}$
  - C  $745 \text{ kJ mol}^{-1}$
  - D  $853 \text{ kJ mol}^{-1}$
  - E  $1181 \text{ kJ mol}^{-1}$
- 25 A chemical equation that represents the reaction of copper(I) phosphide with dilute nitric acid is:



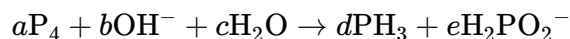
where  $w$ ,  $x$ ,  $y$ , and  $z$  are integers.

What is the value of the sum  $w + x + y + z$ ?

- A 49
- B 53
- C 57
- D 59
- E 63

- 26** When white phosphorus,  $P_4$ , reacts with warm aqueous sodium hydroxide, it undergoes a disproportionation reaction to produce phosphine gas,  $PH_3$ , and hypophosphite ions,  $H_2PO_2^-$ .

The unbalanced ionic equation for this reaction is:



What is the simplest whole-number ratio of  $e : d$  in the balanced equation?

- A** 1 : 1  
**B** 1 : 2  
**C** 2 : 1  
**D** 1 : 3  
**E** 3 : 1  
**F** 1 : 4  
**G** 4 : 1  
**H** 3 : 4
- 27** The apparatus is used to coat a metal spoon with silver. Which of the following statements is/are correct?
1. The spoon should be connected to the negative terminal so that  $Ag^+$  ions are reduced on its surface.
  2. A silver anode can replace  $Ag^+$  ions removed from the electrolyte during plating.
  3. Alternating current is preferred because regularly reversing the electrode polarity increases the thickness of the coating.
- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

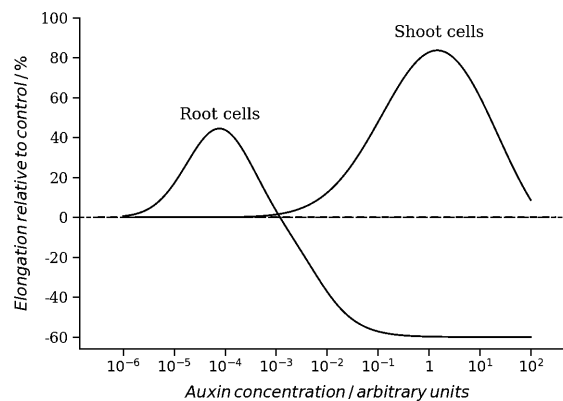
## **Biology**

- 1 The graph shows the effect of auxin concentration on the elongation of root cells and shoot cells, expressed as the percentage change relative to an untreated control.

A germinating seedling is placed horizontally in the dark. Gravity causes auxin to accumulate on the lower side of both the root and the shoot, resulting in an auxin concentration of  $10^{-1}$  arbitrary units on the lower side and  $10^{-4}$  arbitrary units on the upper side in both organs.

Which of the following statements is/are correct?

- 1 The lower side of the shoot elongates faster than its upper side.
- 2 The lower side of the root elongates faster than its upper side.
- 3 The shoot will bend upwards and the root will bend downwards.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 2 A bacterium and a typical animal cell are both undergoing asexual cell division. The bacterium contains a single circular DNA molecule, while the animal cell contains multiple linear chromosomes.

Which row in the table correctly describes the mechanism of genetic material separation in each cell?

|          | <b>Bacterium</b>                                          | <b>Animal Cell</b>                               |
|----------|-----------------------------------------------------------|--------------------------------------------------|
| <b>A</b> | Spindle fibres pull DNA to opposite poles                 | Spindle fibres pull DNA to opposite poles        |
| <b>B</b> | Replicated DNA separates as the cell elongates            | Spindle fibres pull sister chromatids apart      |
| <b>C</b> | Replicated DNA separates as the cell elongates            | Spindle fibres pull homologous chromosomes apart |
| <b>D</b> | Sister chromatids are pulled apart by spindle fibres      | Spindle fibres pull homologous chromosomes apart |
| <b>E</b> | Homologous chromosomes are pulled apart by spindle fibres | Spindle fibres pull sister chromatids apart      |

- 3 During pregnancy in a mammal, exchange of substances occurs between maternal blood and fetal blood across the placenta.

Which row in the table correctly identifies the substances that are present at a higher concentration in the blood of the umbilical vein compared to the blood of the umbilical artery?

Key:

✓ = higher concentration in the umbilical vein

X = not a higher concentration in the umbilical vein

|   | <b>glucose</b> | <b>carbon dioxide</b> | <b>urea</b> |
|---|----------------|-----------------------|-------------|
| A | ✓              | ✓                     | ✓           |
| B | ✓              | ✓                     | X           |
| C | ✓              | X                     | ✓           |
| D | ✓              | X                     | X           |
| E | X              | ✓                     | ✓           |
| F | X              | ✓                     | X           |
| G | X              | X                     | ✓           |
| H | X              | X                     | X           |

- 4 An investigation was carried out to study the rate of protein synthesis in two different strains of a yeast species, Strain X and Strain Y.

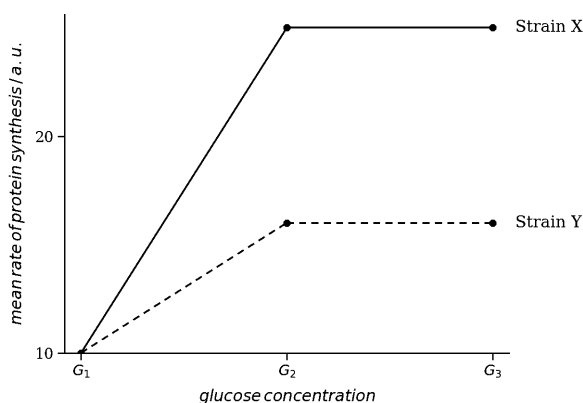
Within each strain, all yeast cells were genetically identical clones.

Samples containing equal numbers of cells from each strain were grown in nutrient solutions with different concentrations of glucose ( $G_1$ ,  $G_2$ , and  $G_3$ , where  $G_1 < G_2 < G_3$ ). All other environmental conditions (temperature, pH, dissolved oxygen, and amino acid concentrations) were kept constant and identical for all cultures.

After a set period of time, the mean rate of protein synthesis per cell was measured for each culture. The results are shown in the graph.

Which of the following statements is/are correct?

- 1 The difference in the mean rate of protein synthesis in Strain X between  $G_1$  and  $G_2$  is due to environmental variation.
- 2 The difference in the mean rate of protein synthesis between Strain X and Strain Y at  $G_2$  is due to genetic variation.
- 3 The failure of the mean rate of protein synthesis to increase in Strain X between  $G_2$  and  $G_3$  could be caused by an environmental factor other than glucose becoming limiting.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

5 Which of the statements about somatic gene therapy used to treat a human genetic disorder caused by a faulty recessive allele are correct?

- 1 A vector is used to introduce a functional copy of the allele into target body cells.
- 2 The introduced allele is incorporated into the genomes of both somatic cells and gametes.
- 3 The functional protein is synthesised using the ribosomes of the patient's own cells.
- 4 Successful treatment ensures that the patient cannot pass the faulty allele to any biological offspring.

- A** 1 and 2 only  
**B** 1 and 3 only  
**C** 2 and 4 only  
**D** 3 and 4 only  
**E** 1, 2 and 3 only  
**F** 1, 3 and 4 only  
**G** 2, 3 and 4 only  
**H** 1, 2, 3 and 4

6 Three different mechanisms of transport across a cell membrane, X, Y and Z, are compared in the table.

| Mechanism of membrane transport | Transports substances across a cell membrane | Requires transmembrane protein carriers or channels | Can result in net movement against a concentration gradient | Requires energy directly from ATP hydrolysis |
|---------------------------------|----------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------|----------------------------------------------|
| X                               | yes                                          | yes                                                 | no                                                          | no                                           |
| Y                               | yes                                          | no                                                  | no                                                          | no                                           |
| Z                               | yes                                          | yes                                                 | yes                                                         | yes                                          |

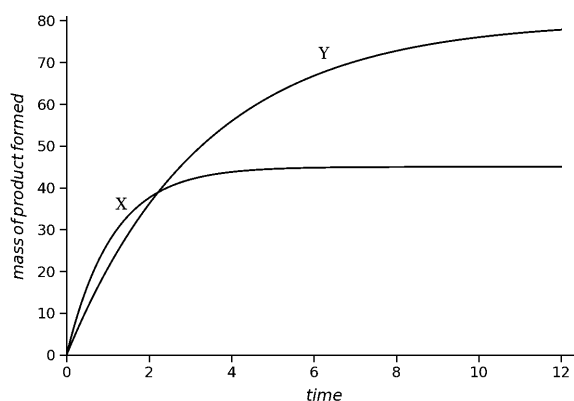
Which row in the table correctly identifies mechanisms X, Y and Z?

|   | X                     | Y                     | Z                     |
|---|-----------------------|-----------------------|-----------------------|
| A | facilitated diffusion | simple diffusion      | active transport      |
| B | facilitated diffusion | active transport      | simple diffusion      |
| C | simple diffusion      | facilitated diffusion | active transport      |
| D | simple diffusion      | active transport      | facilitated diffusion |
| E | active transport      | facilitated diffusion | simple diffusion      |
| F | active transport      | simple diffusion      | facilitated diffusion |

- 7 The graph shows the total mass of product formed over time during an enzyme-catalysed reaction carried out under two different conditions, X and Y.

Which of the following statements could explain the difference between curve X and curve Y?

- 1 Condition X had a higher enzyme concentration and a lower initial mass of substrate than Condition Y.
- 2 Condition X was at a higher temperature that caused progressive thermal denaturation of the enzyme over time, whereas Condition Y was at a lower temperature at which the enzyme remained stable.
- 3 Condition X contained a competitive inhibitor, whereas Condition Y contained no inhibitor.
- 4 Condition X was at a pH further from the enzyme's optimum pH than Condition Y.



- A 1 only
- B 2 only
- C 1 and 2 only
- D 1 and 3 only
- E 2 and 4 only
- F 3 and 4 only
- G 1, 2 and 3 only
- H 1, 2, 3 and 4

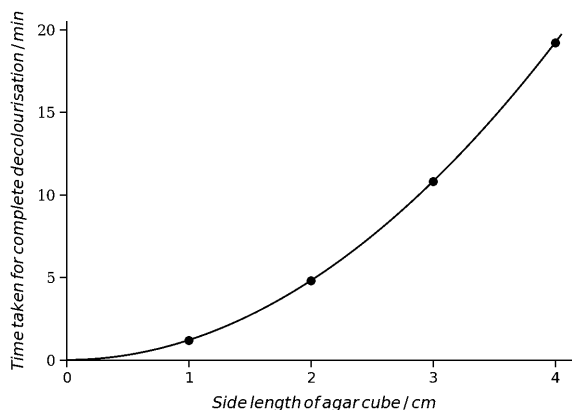
- 8 An investigation was carried out to model the effect of cell size on the rate of exchange of substances by diffusion.

Agar cubes of different side lengths were prepared containing dilute sodium hydroxide and the indicator phenolphthalein (which is pink in alkaline conditions). The cubes were placed into separate beakers containing equal volumes of dilute hydrochloric acid at the same concentration and temperature. The acid diffused into the cubes, neutralising the alkali and causing the cubes to decolourise.

The time taken for each cube to completely decolourise was recorded. The graph shows the results of this experiment.

Which of the following statements about the investigation is/are correct?

- 1 The dependent variable is plotted on the vertical ( $y$ ) axis.
- 2 If the experiment were repeated at a higher temperature, the points plotted would lie below the original curve.
- 3 An agar block with dimensions  $1\text{ cm} \times 2\text{ cm} \times 4\text{ cm}$  would take a longer time to completely decolourise than the  $2\text{ cm} \times 2\text{ cm} \times 2\text{ cm}$  cube.



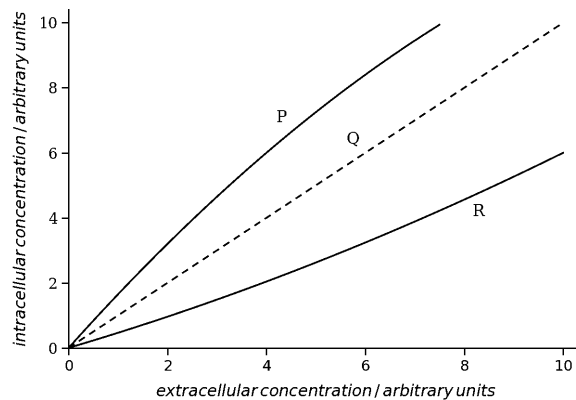
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 9 A cell is placed in solutions containing different concentrations of three uncharged solutes: P, Q, and R. The cell membrane is permeable to all three solutes.

The graph shows the intracellular concentration of each solute once a steady state is reached, as a function of the extracellular concentration. The dashed line represents where intracellular concentration equals extracellular concentration.

Which of the following statements is/are correct?

- 1 Metabolic energy (ATP) is required to maintain the steady-state concentration of solute P.
- 2 Metabolic energy (ATP) is required to maintain the steady-state concentration of solute R.
- 3 If ATP synthesis in the cell is inhibited, the steady-state intracellular concentration of solute R will decrease.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 10** Two mice, both heterozygous for two independently assorting autosomal genes ( $AaBb$ ), are crossed. Under standard Mendelian inheritance with complete dominance, the expected phenotypic ratio among their offspring is 9 : 3 : 3 : 1.

However, across a very large number of offspring from these crosses, four distinct phenotypes are consistently observed in a ratio of 6 : 3 : 2 : 1.

Which of the following statements, if true, could explain this observed phenotypic ratio?

1 Homozygous dominant individuals for gene A ( $AA$ ) die before birth, while  $Aa$  and  $aa$  individuals are viable.

2 Homozygous recessive individuals for gene A ( $aa$ ) die before birth, while  $AA$  and  $Aa$  individuals are viable.

3 Homozygous dominant individuals for gene B ( $BB$ ) die before birth, while  $Bb$  and  $bb$  individuals are viable.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 11** A 20-nucleotide synthetic mRNA molecule is translated in a cell-free translation system. Translation of this mRNA produces a hexapeptide (a peptide consisting of six amino acids) using one continuous reading frame from the 5' to 3' direction without any stop codons.

The sequence of the mRNA is shown below:

5'-AUUGGCAUUGACCGCAUUGC-3'

The table shows the number of each amino acid found in the resulting hexapeptide:

| Amino acid | Number of residues |
|------------|--------------------|
| Leu        | 3                  |
| Pro        | 2                  |
| Met        | 1                  |

What is the primary structure of the hexapeptide from the N-terminus to the C-terminus?

- A** Leu-Pro-Leu-Met-Pro-Leu
- B** Leu-Pro-Met-Leu-Pro-Leu
- C** Pro-Leu-Pro-Met-Leu-Leu
- D** Leu-Leu-Pro-Met-Pro-Leu
- E** Met-Leu-Pro-Leu-Pro-Leu
- F** Met-Pro-Leu-Pro-Leu-Leu
- G** Leu-Pro-Leu-Pro-Met-Leu
- H** Pro-Leu-Leu-Met-Leu-Pro

**12** An experiment was conducted to investigate the gravitropic response of young primary roots.

Three identical batches of seedlings were placed with their primary roots oriented horizontally ( $0^\circ$ ) at time  $t = 0$  h in humid, dark chambers under the following conditions:

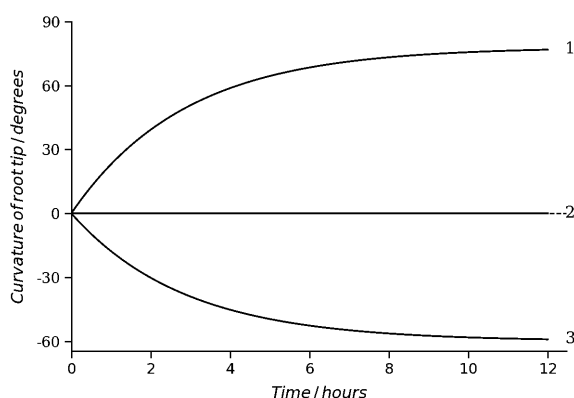
- **Treatment P:** Intact roots with no modifications (control).
- **Treatment Q:** Roots with the root cap surgically removed immediately before placement.
- **Treatment R:** Intact roots with an agar block containing a high concentration of auxin (IAA) placed only on the upper surface of the elongation zone.

The curvature of the root tips was recorded over a period of 12 hours. On the graph:

- a positive angle represents downward bending (towards gravity)
- a negative angle represents upward bending (away from gravity)
- an angle of  $0^\circ$  represents straight horizontal growth.

Which row of the table correctly matches each treatment to its corresponding curve on the graph?

|   | P | Q | R |
|---|---|---|---|
| A | 1 | 2 | 3 |
| B | 1 | 3 | 2 |
| C | 2 | 1 | 3 |
| D | 2 | 3 | 1 |
| E | 3 | 1 | 2 |
| F | 3 | 2 | 1 |
| G | 1 | 2 | 1 |



- 13** An experiment was carried out in total darkness to investigate the gravitropic responses of young seedlings. Three identical, straight seedlings were placed horizontally at time  $t = 0$  h under the following conditions:

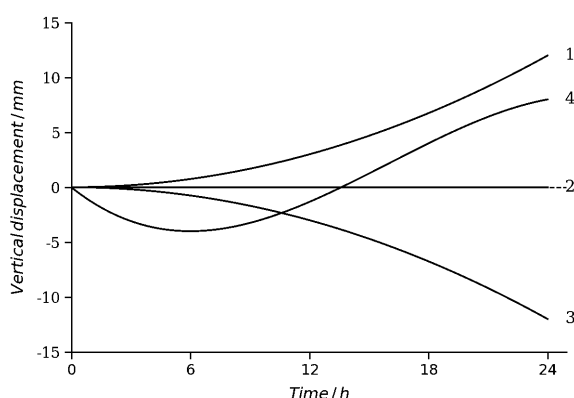
- **P**: An intact seedling kept stationary in a horizontal position; the vertical displacement of the shoot tip from the initial horizontal axis was measured over 24 hours.
- **Q**: An intact seedling kept stationary in a horizontal position; the vertical displacement of the root tip from the initial horizontal axis was measured over 24 hours.
- **R**: An intact seedling placed horizontally on a clinostat rotating slowly and continuously around its horizontal axis; the vertical displacement of the shoot tip from the initial horizontal axis was measured over 24 hours.

For all measurements, an upward displacement from the initial horizontal axis is defined as positive, and a downward displacement is defined as negative.

The graph shows four possible curves (1, 2, 3, and 4) representing vertical displacement over time.

Which row in the table correctly identifies the curves corresponding to seedlings P, Q, and R?

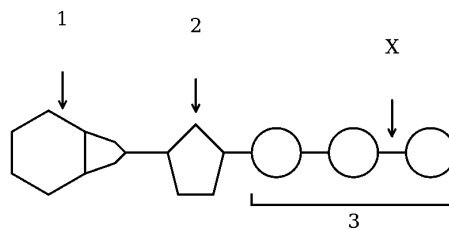
|   | P | Q | R |
|---|---|---|---|
| A | 1 | 2 | 3 |
| B | 1 | 3 | 1 |
| C | 1 | 3 | 2 |
| D | 2 | 1 | 3 |
| E | 2 | 3 | 2 |
| F | 3 | 1 | 2 |
| G | 3 | 1 | 3 |
| H | 3 | 2 | 1 |



- 14** The diagram shows the structure of an adenosine triphosphate (ATP) molecule with components labelled 1, 2, and 3, and a specific bond labelled X.

Which of the following statements is/are correct?

- 1 Components 1 and 2, when bonded to a single phosphate group, form an adenine RNA nucleotide.
- 2 Hydrolysis of bond X produces adenosine diphosphate (ADP) and an inorganic phosphate group ( $P_i$ ).
- 3 Component 2 is a pentose sugar containing one fewer oxygen atom than the pentose sugar found in DNA nucleotides.



*Diagram not to scale*

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 15** The table below shows the mean rates of water uptake by roots and water loss by transpiration for a leafy plant over consecutive 2-hour intervals between 06:00 and 18:00 on a warm day.

| Time interval | Mean rate of water uptake by roots / $\text{cm h}^{-1}$ | Mean rate of water loss by transpiration / $\text{cm h}^{-1}$ |
|---------------|---------------------------------------------------------|---------------------------------------------------------------|
| 06:00 – 08:00 | 15                                                      | 10                                                            |
| 08:00 – 10:00 | 25                                                      | 40                                                            |
| 10:00 – 12:00 | 40                                                      | 65                                                            |
| 12:00 – 14:00 | 50                                                      | 55                                                            |
| 14:00 – 16:00 | 45                                                      | 30                                                            |
| 16:00 – 18:00 | 25                                                      | 10                                                            |

Which of the following statements is/are correct?

- 1 The plant experiences its greatest internal water deficit (highest xylem tension) at 14:00.
- 2 By 18:00, the plant has fully recovered the water deficit accumulated between 08:00 and 14:00.
- 3 The water potential in the leaf mesophyll cells is lower at 14:00 than at 08:00.

- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

- 16** The table shows characteristics of six plant species (A–F) inhabiting a fragmented landscape undergoing rapid climate change.

Which species is most at risk of extinction?

| Species | Generation time | Dispersal ability | Genetic diversity | Ecological niche |
|---------|-----------------|-------------------|-------------------|------------------|
| A       | short           | high              | high              | generalist       |
| B       | short           | low               | high              | specialist       |
| C       | long            | low               | low               | specialist       |
| D       | long            | high              | low               | generalist       |
| E       | long            | low               | high              | generalist       |
| F       | short           | low               | low               | specialist       |

- 17** A segment of double-stranded DNA is transcribed to produce an mRNA molecule containing 600 nucleotides.

The base composition of this mRNA molecule is:

- Adenine (A): 24%
- Uracil (U): 18%
- Cytosine (C): 32%
- Guanine (G): 26%

Which of the following statements about this 600-base-pair DNA segment is/are correct?

- 1 There are 192 cytosine bases in the template DNA strand.
- 2 Thymine accounts for 21% of the total nitrogenous bases in the double-stranded DNA segment.
- 3 There are 1548 hydrogen bonds between the complementary base pairs in this double-stranded DNA segment.

- A** none
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 18** In a species of moth, sex is determined by the ZW sex-chromosome system: females are heterogametic (ZW) and males are homogametic (ZZ).

A breeding experiment begins with a single mated pair of moths (Generation 0).

In each generation:

- each female mates with a male from the same generation and produces a single clutch of 10 offspring;
- the expected sex ratio of male to female offspring in each clutch is 1 : 1, and this ratio is observed in all clutches;
- all offspring survive to maturity and reproduce only within their generation.

What is the expected total number of Z chromosomes present among all individuals in the 4<sup>th</sup> generation of offspring (Generation 4)?

- A** 375
- B** 625
- C** 1250
- D** 1875
- E** 2500
- F** 3125
- G** 3750

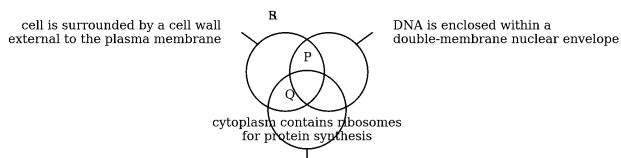
**19** The Venn diagram shows three cellular features used to classify biological cells:

- Feature 1: Cell is surrounded by a cell wall external to the plasma membrane
- Feature 2: DNA is enclosed within a double-membrane nuclear envelope
- Feature 3: Cytoplasm contains ribosomes for protein synthesis

A bacterial cell, a yeast cell, and a human cheek epithelial cell can each be placed into one of the labelled regions P, Q, R, or S.

Which row in the table correctly identifies the region for each cell type?

|   | bacterial cell | yeast cell | human cheek epithelial cell |
|---|----------------|------------|-----------------------------|
| A | Q              | S          | R                           |
| B | Q              | R          | P                           |
| C | P              | S          | R                           |
| D | S              | Q          | P                           |
| E | S              | S          | R                           |
| F | R              | S          | Q                           |
| G | Q              | P          | R                           |
| H | P              | Q          | S                           |



**20** Root hair cells of a plant are placed in an aerated nutrient solution.

The concentrations of two dissolved ions, ion X and ion Y, and the water potential in the solution and in the root hair cell cytoplasm are:

- Ion X: concentration in solution =  $0.1 \text{ mmol dm}^{-3}$ ; concentration in cytoplasm =  $4.5 \text{ mmol dm}^{-3}$
- Ion Y: concentration in solution =  $8.0 \text{ mmol dm}^{-3}$ ; concentration in cytoplasm =  $1.2 \text{ mmol dm}^{-3}$
- Water potential: solution =  $-80 \text{ kPa}$ ; cytoplasm =  $-450 \text{ kPa}$

Under normal aerated conditions, there is net movement of ion X, ion Y, and water into the root hair cells.

A chemical that specifically inhibits ATP synthesis is added to the nutrient solution.

Which row in the table shows the net movement of ion X, ion Y, and water across the root hair cell surface membrane immediately after the chemical is added?

|   | net movement of ion X       | net movement of ion Y       | net movement of water |
|---|-----------------------------|-----------------------------|-----------------------|
| A | moves into the cell         | moves into the cell         | moves into the cell   |
| B | moves into the cell         | moves into the cell         | moves out of the cell |
| C | moves into the cell         | does not move into the cell | moves into the cell   |
| D | moves into the cell         | does not move into the cell | moves out of the cell |
| E | does not move into the cell | moves into the cell         | moves into the cell   |
| F | does not move into the cell | moves into the cell         | moves out of the cell |
| G | does not move into the cell | does not move into the cell | moves into the cell   |
| H | does not move into the cell | does not move into the cell | moves out of the cell |

- 21** The table shows several mRNA codons and the amino acids or translation functions they code for. All other codons not listed code for amino acids other than glutamic acid, aspartic acid, lysine, glutamine, or tryptophan, and none of the unlisted codons act as stop codons.

| mRNA codon | Amino acid / function |
|------------|-----------------------|
| GAA        | glutamic acid         |
| GAG        | glutamic acid         |
| GAC        | aspartic acid         |
| GAU        | aspartic acid         |
| AAG        | lysine                |
| CAG        | glutamine             |
| UAG        | STOP                  |
| UGA        | STOP                  |
| UAA        | STOP                  |
| UGG        | tryptophan            |

Consider the following six-base segment of an mRNA molecule:

5'-GAG UGG-3'

A single random base substitution mutation occurs within this six-base sequence. The mutation is equally likely to occur at any of the six nucleotide positions. At the mutated position, the original base is replaced by one of the other three RNA bases with equal probability ( $\frac{1}{3}$  each).

Assuming no other mutations occur, what are the probabilities of the following outcomes?

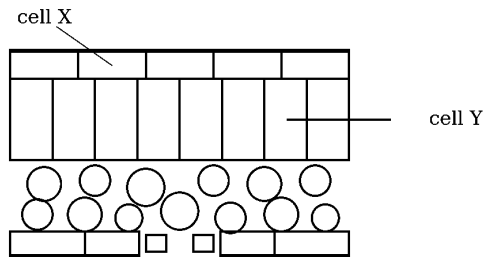
|   | Probability that the mutation is silent (both codons still code for glutamic acid followed by tryptophan) | Probability that the mutation produces a stop codon in this sequence |
|---|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| A | $\frac{118}{1000}$                                                                                        | $\frac{118}{1000}$                                                   |
| B | $\frac{118}{1000}$                                                                                        | $\frac{1}{1000}$                                                     |
| C | $\frac{118}{1000}$                                                                                        | $\frac{1}{1000}$                                                     |
| D | $\frac{118}{1000}$                                                                                        | $\frac{1}{1000}$                                                     |
| E | $\frac{1}{1000}$                                                                                          | $\frac{118}{1000}$                                                   |
| F | $\frac{1}{1000}$                                                                                          | $\frac{1}{1000}$                                                     |
| G | $\frac{1}{1000}$                                                                                          | $\frac{1}{1000}$                                                     |
| H | $\frac{1}{1000}$                                                                                          | $\frac{1}{1000}$                                                     |

**22** The diagram shows a simplified cross-section through part of a dicotyledonous leaf.

Consider the following statements about the two cells labelled X and Y:

- 1 Both cell X and cell Y belong to the same plant tissue.
- 2 Both cell X and cell Y contain the same nuclear genes.
- 3 Both cell X and cell Y contain chloroplasts.

Which of these statements is/are correct?



*Diagram not to scale*

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 23** The table below displays annual carbon flux data for three different ecosystems, measured in arbitrary units of mass per area per year.

| Ecosystem | Gross Primary Production (GPP) | Autotrophic Respiration (Ra) | Heterotrophic Respiration (Rh) |
|-----------|--------------------------------|------------------------------|--------------------------------|
| 1         | 120                            | 50                           | 40                             |
| 2         | 100                            | 40                           | 65                             |
| 3         | 150                            | 80                           | 70                             |

Which of the following conclusions regarding the carbon balance of these ecosystems is supported by the data?

- A** Ecosystem 3 acts as the largest net carbon sink because it has the highest Gross Primary Production.
- B** Ecosystem 2 acts as a net carbon source because its total respiration exceeds its Gross Primary Production.
- C** Ecosystems 1 and 3 contribute equally to the net removal of atmospheric carbon because their Net Primary Production is identical.
- D** Ecosystem 2 acts as a net carbon sink because the carbon fixed by photosynthesis exceeds the carbon released by producers.
- E** Ecosystem 1 is a net carbon source because the sum of autotrophic and heterotrophic respiration is highest relative to its GPP.

- 24** A marine biologist investigated the population of a burrowing bivalve species on a large intertidal mudflat.

A square core sampler with internal base dimensions of  $0.20 \text{ m} \times 0.20 \text{ m}$  was pushed into the sediment to a uniform depth of  $0.15 \text{ m}$ . The sediment from the core was sieved to count the number of live bivalves present. This procedure was repeated 10 times.

Due to deep, soft mud further out on the mudflat, all 10 core samples were taken within  $2 \text{ m}$  of the high-tide line along the upper shore.

The number of bivalves recorded in each core is shown in the table:

| Sample number      | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|--------------------|---|---|---|---|---|---|---|---|---|----|
| Number of bivalves | 3 | 7 | 0 | 5 | 0 | 4 | 8 | 2 | 6 | 5  |

The biologist made the following statements:

- 1 Each core sampled a sediment volume of  $6.0 \times 10^{-3} \text{ m}^3$ .
- 2 The frequency of occurrence of the bivalves in the samples was 80%.
- 3 An accurate estimate of the total bivalve population on the entire mudflat could be calculated using these data if the total surface area of the mudflat was known.

Which of the statements about the investigation is/are correct?

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

**25** In domestic cats, fur colour is controlled by an X-linked gene with two codominant alleles:

- \* allele  $X^B$  produces black fur
- \* allele  $X^O$  produces orange fur
- \* heterozygous females ( $X^B X^O$ ) have tortoiseshell fur

Two tortoiseshell female cats each mate with a different male cat:

- \* Female 1 mates with a black male cat ( $X^B Y$ )
- \* Female 2 mates with an orange male cat ( $X^O Y$ )

Each mating produces an equal number of male and female offspring, and both litters contain the same total number of kittens.

What is the expected proportion of all female kittens that are tortoiseshell, and what is the expected proportion of all kittens (males and females combined) across both litters that are orange?

|   | expected proportion of female kittens that are tortoiseshell | expected proportion of all kittens that are orange |
|---|--------------------------------------------------------------|----------------------------------------------------|
| A | 0.25                                                         | 0.250                                              |
| B | 0.25                                                         | 0.375                                              |
| C | 0.25                                                         | 0.500                                              |
| D | 0.50                                                         | 0.250                                              |
| E | 0.50                                                         | 0.375                                              |
| F | 0.50                                                         | 0.500                                              |
| G | 0.75                                                         | 0.375                                              |
| H | 0.75                                                         | 0.500                                              |

- 26** A linear double-stranded DNA molecule contains 3000 base pairs and has a total of 7800 hydrogen bonds between its paired bases.

Which of the following statements is/are correct?

- 1 Guanine accounts for 60% of the bases in this DNA molecule.
- 2 There are 1200 adenine bases present in this DNA molecule.
- 3 A single strand of this DNA molecule contains 2999 phosphodiester bonds.

- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

- 27** The table shows the partial pressures of oxygen ( $pO_2$ ) and carbon dioxide ( $pCO_2$ ), and the concentration of urea, in blood entering and leaving three different organs (X, Y, and Z) of a healthy human at rest:

| Organ | $pO_2$ entering / kPa | $pO_2$ leaving / kPa | $pCO_2$ entering / kPa | $pCO_2$ leaving / kPa | Urea entering / mmol dm <sup>-3</sup> | Urea leaving / mmol dm <sup>-3</sup> |
|-------|-----------------------|----------------------|------------------------|-----------------------|---------------------------------------|--------------------------------------|
| X     | 13.0                  | 5.5                  | 5.2                    | 6.0                   | 5.0                                   | 2.5                                  |
| Y     | 5.5                   | 13.0                 | 6.0                    | 5.2                   | 5.0                                   | 5.0                                  |
| Z     | 13.0                  | 5.5                  | 5.2                    | 6.0                   | 5.0                                   | 7.5                                  |

Which row in the table below correctly identifies organs X, Y, and Z?

|   | Organ X | Organ Y | Organ Z |
|---|---------|---------|---------|
| A | kidney  | lungs   | liver   |
| B | kidney  | liver   | lungs   |
| C | lungs   | kidney  | liver   |
| D | lungs   | liver   | kidney  |
| E | liver   | kidney  | lungs   |
| F | liver   | lungs   | kidney  |